



THE UNIVERSITY OF LAHORE

Department of CS&IT

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LAB_TASK (week-8)

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Name: Ayan Ahmad

SAP ID: 70147501

Class / Section: BSCS-6C

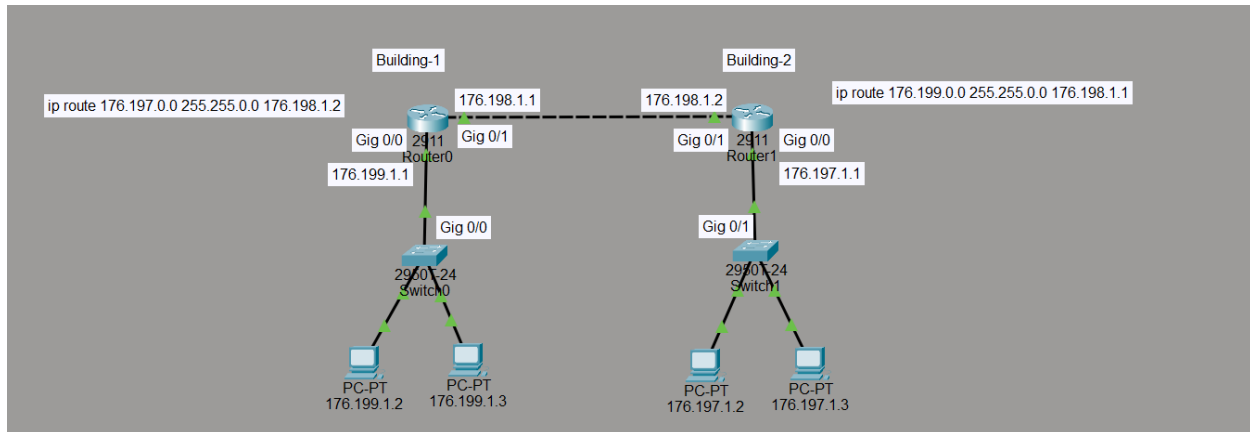
Submitted To: Sir Muhammad Shahzad

Subject: Computer Networks

Lab Task

STATIC ROUTING:

Topology Pic:



Building-1

Router 0:

Information:

Device Name: Router0
Device Model: 2911
Hostname: Router

Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
GigabitEthernet0/0	Up	--	176.199.1.1/16	<not set>	0001.C7E5.0101
GigabitEthernet0/1	Up	--	176.198.1.1/16	<not set>	0001.C7E5.0102
GigabitEthernet0/2	Down	--	<not set>	<not set>	0001.C7E5.0103
Vlan1	Down	1	<not set>	<not set>	0003.E4B8.60DA

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router0

Configuration:

Router>en

Router#show run

Building configuration...

Current configuration : 761 bytes

!

version 15.1

no service timestamps log datetime msec

no service timestamps debug datetime msec

Lab Task

```
no service password-encryption
!
hostname Router
ip cef
no ipv6 cef
!
license udi pid CISCO2911/K9 sn FTX152493A0-
!
spanning-tree mode pvst
!
interface GigabitEthernet0/0
ip address 176.199.1.1 255.255.0.0
duplex auto
speed auto
!
interface GigabitEthernet0/1
ip address 176.198.1.1 255.255.0.0
duplex auto
speed auto
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Vlan1
no ip address
shutdown
!
ip classless
```

Lab Task

```
ip route 176.197.0.0 255.255.0.0 176.198.1.2
```

```
!
```

```
ip flow-export version 9
```

```
!
```

```
line con 0
```

```
!
```

```
line aux 0
```

```
!
```

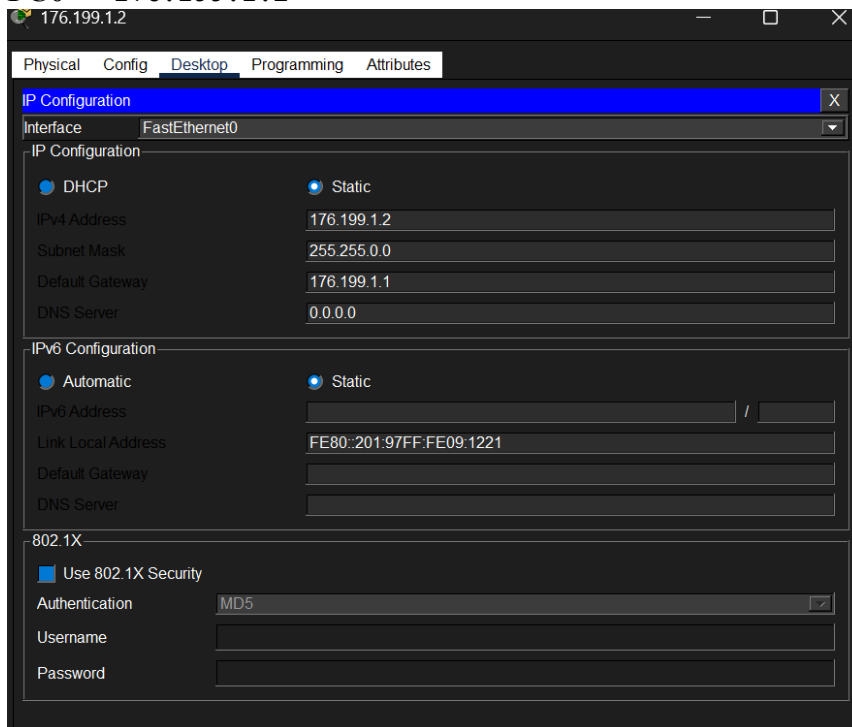
```
line vty 0 4
```

```
login
```

```
!end
```

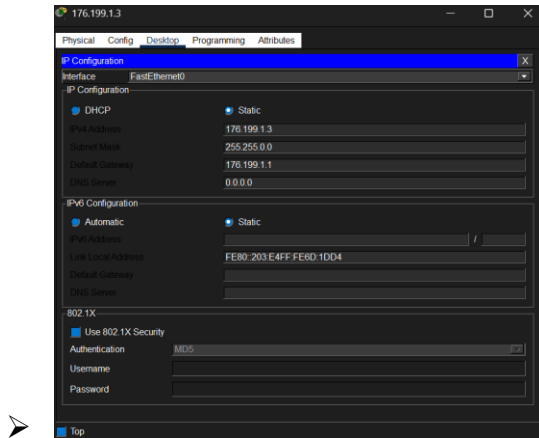
Switch0 → 2960-24 :

➤ **PC0 → 176.199.1.2**



➤ **PC1 → 176.199.1.3**

Lab Task



Building-2:

Router 1:

Information:

Device Name: Router1					
Device Model: 2911					
Hostname: Router					
Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
GigabitEthernet0/0	Up	--	176.197.1.1/16	<not set>	0060.3EE8.B001
GigabitEthernet0/1	Up	--	176.198.1.2/16	<not set>	0060.3EE8.B002
GigabitEthernet0/2	Down	--	<not set>	<not set>	0060.3EE8.B003
Vlan1	Down	1	<not set>	<not set>	00D0.97D6.B594
Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Router1					

Configuration:

Router>en

Router#show run

Building configuration...

Current configuration : 761 bytes

!

version 15.1

no service timestamps log datetime msec

no service timestamps debug datetime msec

no service password-encryption

!

hostname Router

ip cef

no ipv6 cef

!

Lab Task

```
license udi pid CISCO2911/K9 sn FTX1524DID4-
```

```
!
```

```
spanning-tree mode pvst
```

```
!
```

```
!
```

```
interface GigabitEthernet0/0
```

```
ip address 176.197.1.1 255.255.0.0
```

```
duplex auto
```

```
speed auto
```

```
!
```

```
interface GigabitEthernet0/1
```

```
ip address 176.198.1.2 255.255.0.0
```

```
duplex auto
```

```
speed auto
```

```
!
```

```
interface GigabitEthernet0/2
```

```
no ip address
```

```
duplex auto
```

```
speed auto
```

```
shutdown
```

```
!
```

```
interface Vlan1
```

```
no ip address
```

```
shutdown
```

```
!
```

```
ip classless
```

```
ip route 176.199.0.0 255.255.0.0 176.198.1.1
```

```
!
```

```
ip flow-export version 9
```

```
!
```

```
line con 0
```

```
!
```

```
line aux 0
```

```
!
```

```
line vty 0 4
```

```
login
```

Lab Task

end

Switch1 → 2960-24:

➤ **PC2 → 176.197.1.2**

The screenshot shows the configuration window for PC2 with IP address 176.197.1.2. The 'Desktop' tab is selected. Under 'IP Configuration', the 'Static' radio button is chosen. The IPv4 Address is 176.197.1.2, Subnet Mask is 255.255.0.0, Default Gateway is 176.197.1.1, and DNS Server is 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is chosen. The Link Local Address is FE80::202:17FF:FE97:8DC6. Under '802.1X', the 'Use 802.1X Security' checkbox is checked, and the Authentication is set to MD5. A 'Top' button is at the bottom left.

Configuration Section	Option / Value
IP Configuration	Static (Selected)
IPv4 Address	176.197.1.2
Subnet Mask	255.255.0.0
Default Gateway	176.197.1.1
DNS Server	0.0.0.0
IPv6 Configuration	Static (Selected)
Link Local Address	FE80::202:17FF:FE97:8DC6
802.1X	Use 802.1X Security (Checked)
Authentication	MD5

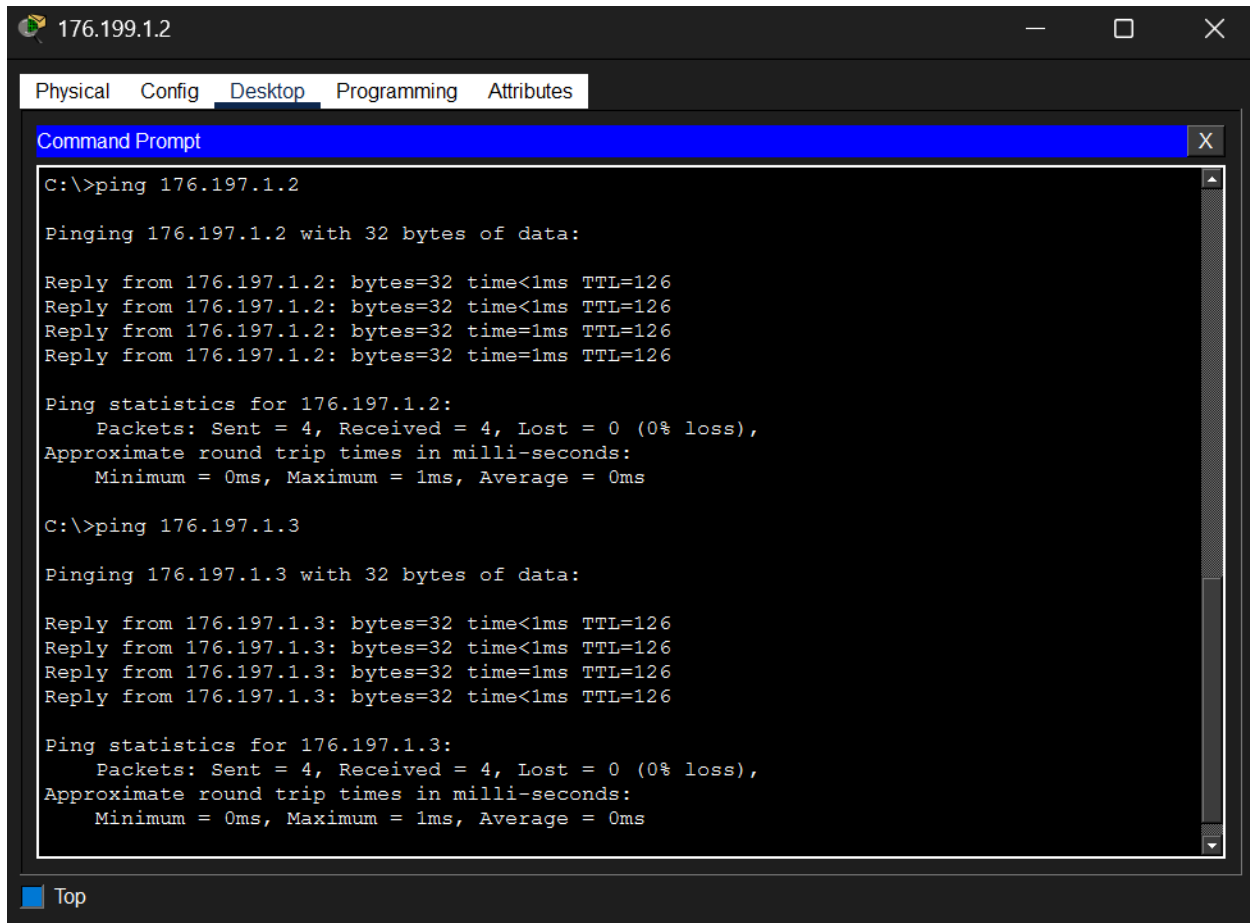
➤ **PC3 → 176.197.1.3**

The screenshot shows the configuration window for PC3 with IP address 176.197.1.3. The 'Desktop' tab is selected. Under 'IP Configuration', the 'Static' radio button is chosen. The IPv4 Address is 176.197.1.3, Subnet Mask is 255.255.0.0, Default Gateway is 176.197.1.1, and DNS Server is 0.0.0.0. Under 'IPv6 Configuration', the 'Static' radio button is chosen. The Link Local Address is FE80::250:FFF:FE91:EDC. Under '802.1X', the 'Use 802.1X Security' checkbox is checked, and the Authentication is set to MD5. A 'Top' button is at the bottom left.

Configuration Section	Option / Value
IP Configuration	Static (Selected)
IPv4 Address	176.197.1.3
Subnet Mask	255.255.0.0
Default Gateway	176.197.1.1
DNS Server	0.0.0.0
IPv6 Configuration	Static (Selected)
Link Local Address	FE80::250:FFF:FE91:EDC
802.1X	Use 802.1X Security (Checked)
Authentication	MD5

PING Commands

Building 1 with both pc of building 2:



The screenshot shows a network configuration window for the IP address 176.199.1.2. The window has tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a Command Prompt window. The Command Prompt shows the execution of two ping commands. The first command is 'ping 176.197.1.2', which returns four successful replies with 32 bytes of data, a time of less than 1ms, and a TTL of 126. The second command is 'ping 176.197.1.3', which also returns four successful replies with 32 bytes of data, a time of less than 1ms, and a TTL of 126. Ping statistics for both destinations show 4 packets sent, 4 received, and 0% loss, with round trip times of 0ms.

```
C:\>ping 176.197.1.2

Pinging 176.197.1.2 with 32 bytes of data:

Reply from 176.197.1.2: bytes=32 time<1ms TTL=126
Reply from 176.197.1.2: bytes=32 time<1ms TTL=126
Reply from 176.197.1.2: bytes=32 time=1ms TTL=126
Reply from 176.197.1.2: bytes=32 time=1ms TTL=126

Ping statistics for 176.197.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 176.197.1.3

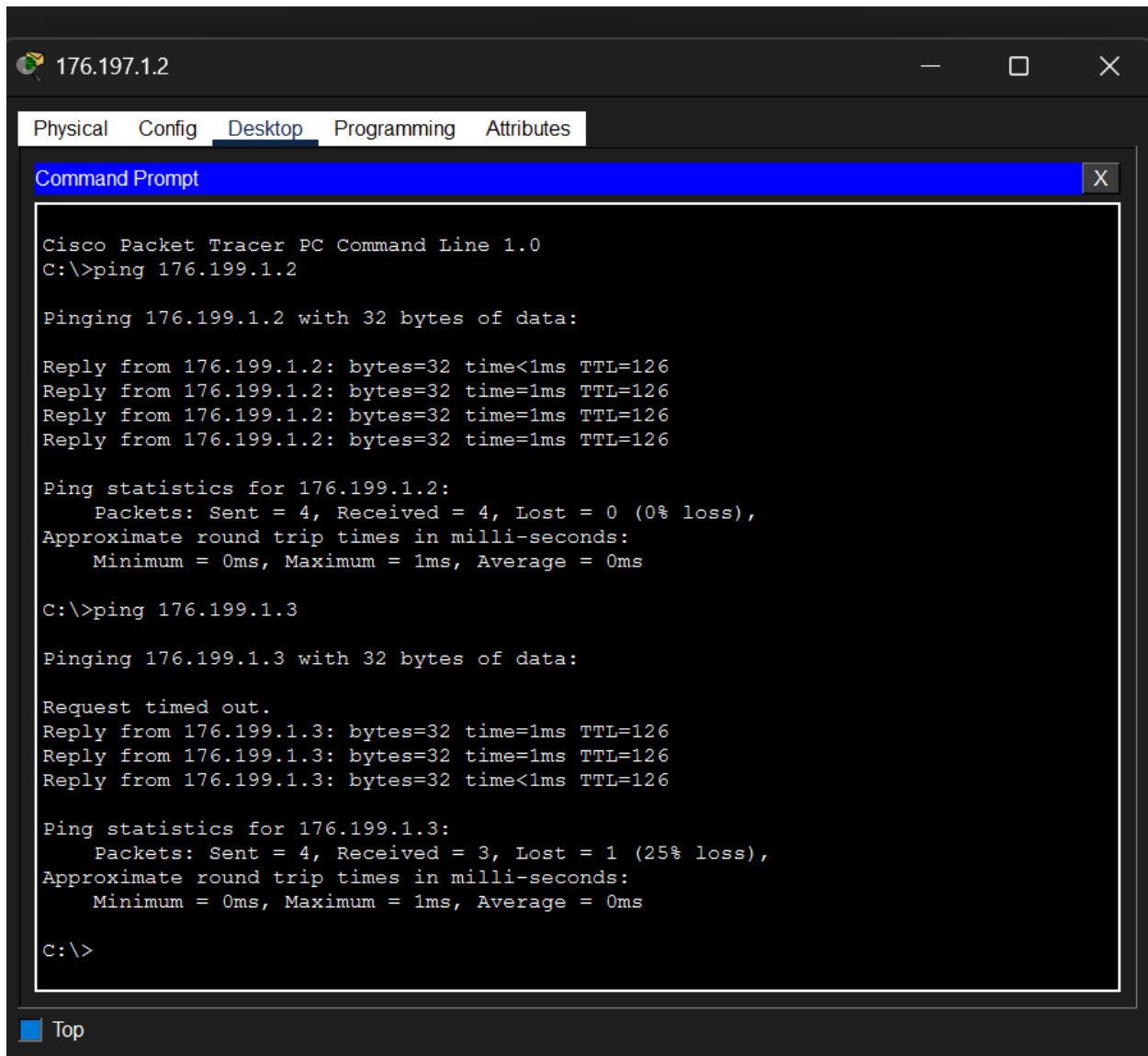
Pinging 176.197.1.3 with 32 bytes of data:

Reply from 176.197.1.3: bytes=32 time<1ms TTL=126
Reply from 176.197.1.3: bytes=32 time<1ms TTL=126
Reply from 176.197.1.3: bytes=32 time=1ms TTL=126
Reply from 176.197.1.3: bytes=32 time<1ms TTL=126

Ping statistics for 176.197.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Building 2 with both pc of building 1:

Lab Task



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 176.199.1.2

Pinging 176.199.1.2 with 32 bytes of data:

Reply from 176.199.1.2: bytes=32 time<1ms TTL=126
Reply from 176.199.1.2: bytes=32 time=1ms TTL=126
Reply from 176.199.1.2: bytes=32 time=1ms TTL=126
Reply from 176.199.1.2: bytes=32 time=1ms TTL=126

Ping statistics for 176.199.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 176.199.1.3

Pinging 176.199.1.3 with 32 bytes of data:

Request timed out.
Reply from 176.199.1.3: bytes=32 time=1ms TTL=126
Reply from 176.199.1.3: bytes=32 time=1ms TTL=126
Reply from 176.199.1.3: bytes=32 time<1ms TTL=126

Ping statistics for 176.199.1.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Top